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(57) Abstract :
The present invention relates to the investigation of temperature-dependent electrical transport properties of Au/$SiO_2$/n-GaN Metal–Insulator–Semiconductor (MIS) Schottky diodes over a wide temperature range of 120 K to 390 K. The electrical characteristics of the MIS Schottky diode are evaluated using temperature-dependent current–voltage (I–V) and capacitance–voltage (C–V) measurements. The barrier height, ideality factor, series resistance, interface state density and other electrical parameters are determined as a function of temperature. The barrier height increases and the ideality factor decreases with increasing temperature, while the series resistance and interface state density decrease with increasing temperature. The observed temperature dependence is associated with the electrical transport behavior and barrier inhomogeneities at the metal–insulator–semiconductor interface. The invention provides a method for evaluating the electrical characteristics and temperature-dependent performance of Au/$SiO_2$/n-GaN MIS Schottky diodes for wide-temperature semiconductor applications.

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# Description

1. **Abstract**

The present invention relates to the investigation of temperature-dependent electrical transport properties of Au/$SiO_2$/n-GaN Metal–Insulator–Semiconductor (MIS) Schottky diodes over a wide temperature range of 120 K to 390 K. The electrical characteristics of the MIS Schottky diode are evaluated using temperature-dependent current–voltage (I–V) and capacitance–voltage (C–V) measurements. The barrier height, ideality factor, series resistance, interface state density and other electrical parameters are determined as a function of temperature. The barrier height increases and the ideality factor decreases with increasing temperature, while the series resistance and interface state density decrease with increasing temperature. The observed temperature dependence is associated with the electrical transport behavior and barrier inhomogeneities at the metal–insulator–semiconductor interface. The invention provides a method for evaluating the electrical characteristics and temperature-dependent performance of Au/$SiO_2$/n-GaN MIS Schottky diodes for wide-temperature semiconductor applications.

## DESCRIPTION

2. **Title**: Electrical Transport Characteristics of Au/$SiO_2$/n-GaN MIS Schottky Diodes across a Wide Temperature Range

3. **Field of Invention:**

The present invention relates to semiconductor devices**,** particularly to the temperature-dependent electrical characterization and evaluation of Au/$SiO_2$/n-GaN Metal–Insulator–Semiconductor (MIS) Schottky diodes over a wide temperature range.

4. **Background of Invention:**

Metal–Insulator–Semiconductor (MIS) Schottky structures are useful semiconductor devices whose electrical characteristics depend on the properties of the metal–insulator–semiconductor interfaces. The electrical transport characteristics of such

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structures can vary significantly with temperature due to changes in barrier height, ideality factor, series resistance and interface state density.

Therefore, studying the electrical characteristics over a wide temperature range is useful for understanding the temperature-dependent transport behavior and performance of MIS Schottky diodes.

The present invention investigates the electrical transport properties of an Au/$SiO_2$/n-GaN MIS Schottky diode over the temperature range of 120 K to 390 K using temperature-dependent I–V and C–V measurements. The reported measurements show strong temperature dependence of barrier height, ideality factor, series resistance and interface state density.

## 5. Objective of Invention:

The main objective of the present invention is to evaluate the temperature-dependent electrical transport properties of Au/$SiO_2$/n-GaN MIS Schottky diodes over the temperature range of 120 K to 390 K**.**

A further objective is to determine the variation of barrier height, ideality factor, series resistance and interface state density with temperature.

## 6. Summary of Invention:

The present invention provides a temperature-dependent electrical characterization of an Au/$SiO_2$/n-GaN MIS Schottky diode over a temperature range of 120 K to 390 K**.**

The MIS Schottky diode is characterized using temperature-dependent I–V and C–V measurements. The measured characteristics are used to determine the barrier height, ideality factor, series resistance and interface state density.

The results demonstrate that the zero-bias barrier height increases and the ideality factor decreases with increasing temperature. The series resistance obtained using Cheung's method decreases with increasing temperature. The interface state density also decreases with increasing temperature.

The temperature-dependent behavior of the electrical parameters provides information regarding the transport characteristics and interface properties of the Au/$SiO_2$/n-GaN MIS Schottky diode.

**7. Information about Analysis:**

The temperature-dependent I–V and C–V characteristics of the Au/$SiO_2$/n-GaN MIS Schottky diode were analyzed over the temperature range of 120 K to 390 K**.** The barrier height and ideality factor were determined from the I–V characteristics, while the series resistance was obtained using Cheung's method. The barrier height increased from 0.35 eV at 120 K to 1.02 eV at 390 K**,** whereas the ideality factor decreased from 3.63 to 1.36**.** The series resistance and interface state density were observed to decrease with increasing temperature.

The temperature-dependent C–V characteristics were additionally analyzed to determine the barrier height and donor concentration. The donor concentration was found to vary from approximately $1.67 \times 10^{17}$ $cm^{-3}$ at 120 K to $2.06 \times 10^{17}$ $cm^{-3}$ at 390 K**.**

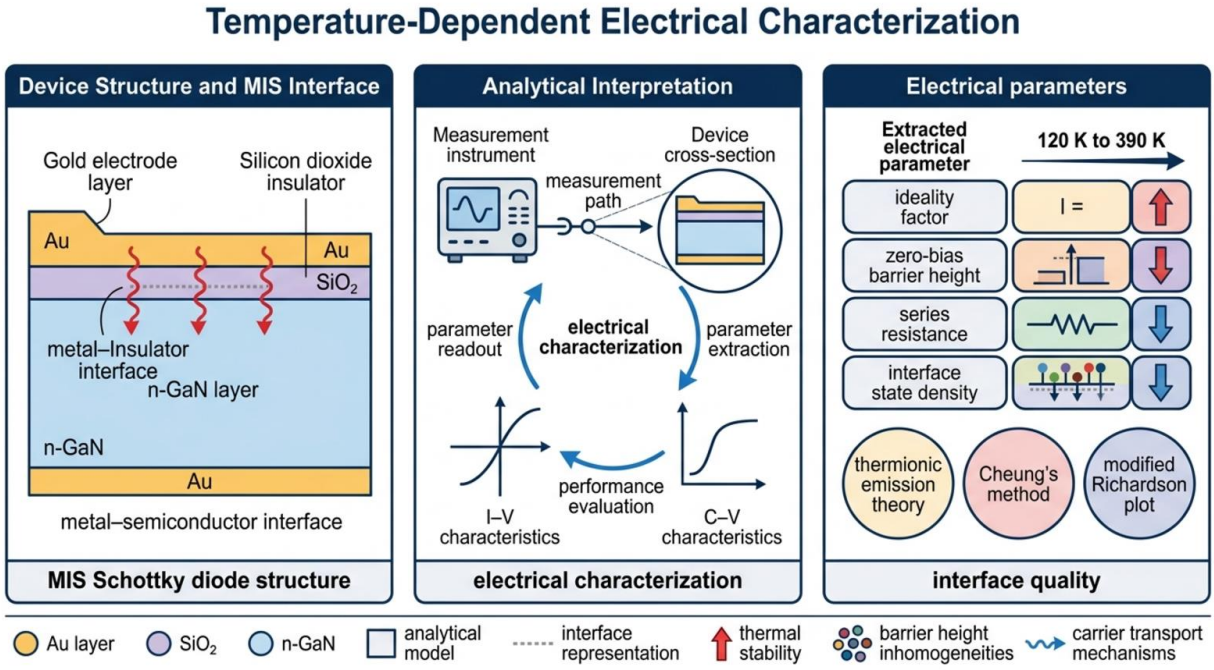
**8. Best Method:**

We concluded that temperature-dependent electrical characterization of the Au/$SiO_2$/n-GaN MIS Schottky diode over the temperature range of 120 K to 390 K provides an effective method for evaluating its electrical transport properties. The combined analysis of I–V and C–V characteristics enables determination of barrier height, ideality factor, series resistance and interface state density, thereby providing a comprehensive evaluation of the temperature-dependent performance of the MIS structure.

**9. Industrial Applicability:**

The present invention is applicable for the characterization and evaluation of Au/$SiO_2$/n-GaN MIS Schottky diodes over a wide temperature range of 120 K to 390 K. The temperature-dependent electrical analysis can be used for evaluating

semiconductor devices intended for wide-temperature, high-power and high-frequency electronic applications.




## CLAIMS:

**Claim 1.** A temperature-dependent electrical characterization method for an Au/$SiO_2$/n-GaN Metal–Insulator–Semiconductor (MIS) Schottky diode, comprising measuring current–voltage (I–V) and capacitance–voltage (C–V) characteristics over a temperature range of **120 K to 390 K**.

**Claim 2.** The method according to Claim 1, wherein the temperature-dependent I–V characteristics are used to determine the **barrier height and ideality factor** of the MIS Schottky diode.

**Claim 3.** The method according to Claim 1, wherein the **series resistance** is determined from the forward-bias I–V characteristics using Cheung's method.

**Claim 4.** The method according to Claim 1, wherein the temperature-dependent C–V characteristics are used to determine the **barrier height and donor concentration** of the MIS Schottky diode.

**Claim 5.** The method according to Claim 1, wherein the **interface state density is evaluated as a function of temperature** over the range of 120 K to 390 K.

**Claim 6.** The method according to Claim 1, wherein the temperature-dependent electrical parameters are used for evaluating the **electrical transport properties and temperature-dependent performance** of the Au/$SiO_2$/n-GaN MIS Schottky diode.